

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	Rising Waters: AI-Powered Flood Risk Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Flood Risk Prediction	Predicts flood risk level (Low, Moderate, High) using rainfall, river level, land use, and other hydrological parameters.	ML Prediction Engine	Done	Core Feature
2	Hydrological Data Input	Accepts user input through an intuitive web interface for rainfall, river level, soil type, land use, and other relevant data.	User Interface	Done	High
3	Machine Learning Prediction Model	Uses the trained machine learning model to generate accurate flood risk predictions.	Machine Learning	Done	Core Feature
4	Risk Probability Display	Displays probability (in %) of flood occurrence along with the predicted risk level to improve user awareness.	Prediction Module	Done	Medium
5	Flood Data Visualization	Presents rainfall trends, river levels, and risk distribution using interactive charts and graphs.	Analytics Dashboard	Done	High
6	Data Preprocessing	Cleans, validates, and transforms hydrological data before feeding it to the machine learning model.	Data Processing	Done	High

7	Model Performance Evaluation	Evaluates the model using Accuracy, Confusion Matrix, and Classification Report for reliable predictions.	Evaluation Module	Done	Medium
8	Responsive Web Application	Provides an easy-to-use interface accessible from desktop and mobile browsers.	Frontend (Flask UI)	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Flood Risk Prediction, ML Prediction, Data Input, Data Preprocessing, Model Evaluation)
Additional / Nice-to-Have Features	Risk Probability Display, Flood Data Visualization
Features Tested & Verified	Flood Data Input, Flood Risk Prediction, Risk Probability Analysis, ML Prediction, Prediction Dashboard

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Hydrological data input form, Risk prediction result dashboard
2	Backend / Logic	3	Flood risk prediction, Risk probability analysis, Input validation
3	Database / Storage	1	Hydrological dataset (CSV) and trained ML model
4	API / Integration	1	Flask application connecting frontend with the prediction model
5	Security / Authentication	1	Input Validation and Secure Request Handling